# A Two-Stage Perceptual Account of Interval Consonance

David De Roure

Working note, June 2026

**Abstract**

Euler's *Gradus Suavitatis* (1739) assigns a dissonance value to a musical interval $p/q$ by the formula $\mathrm{G}(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, where $\Omega^*(n) = \sum_i e_i(p_i - 1)$ sums the weighted prime exponents of $n$. We show that the simpler asymmetric formula $f(p/q) = p + \Omega^*(q)$ achieves a higher rank correlation with human consonance ratings ($\rho = 0.989$ vs. $0.979$), and that this formula admits a natural two-stage perceptual interpretation grounded in harmonic series theory. We also show that Gradus is exactly equivalent to a weighted harmonic coincidence count with weights $w(n) = \Omega^*(n)$, and that the simpler metric $\max(p, q)$ — equivalent to Galileo's pulse-coincidence model — achieves the same correlation as $f(p/q)$.

## 1 Background

Euler's Gradus Suavitatis, catalogued in OEIS A275314, assigns to each musical interval a positive integer measuring its arithmetic dissonance. For a ratio $p/q$ in lowest terms, the characteristic number is $n = p \cdot q$, and:

$$\mathrm{G}(p/q) = 1 + \Omega^*(p \cdot q)$$

where $\Omega^*$ is the completely additive function $\Omega^*(n) = \sum_i e_i(p_i - 1)$. Since $\Omega^*$ is completely additive, $\mathrm{G}(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, treating numerator and denominator symmetrically. The weights $(p_i - 1)$ for prime $p_i$ encode Euler's aesthetic judgment that large primes represent greater "difficulty."

Euler's formula correctly ranks the 13 standard intervals of Western music against human consonance ratings [7] with Spearman $\rho = 0.979$, and is exactly tied only on the major third/major sixth pair and on a three-way tie at Gradus 8.

## 2 Gradus as Harmonic Coincidence

For two notes at frequency ratio $p : q$ (coprime), exact harmonic coincidences occur at harmonics $p, 2p, 3p, \ldots$ of the lower note. For a fixed count $M$ of coincidences, the weighted sum with weights $w(n) = \Omega^*(n)$ yields:

$$\sum_{k=1}^{M} \Omega^*(k \cdot p) \;=\; \Omega^*(p) \cdot M + C(M)$$

where $C(M) = \sum_{k=1}^{M} \Omega^*(k)$ is a constant independent of $p$. This follows from the complete additivity of $\Omega^*$: $\Omega^*(k \cdot p) = \Omega^*(k) + \Omega^*(p)$. The symmetric score over both notes gives:

$$\text{Score} = M \cdot (\mathrm{G}(p/q) - 1) + 2\,C(M)$$

**Gradus is therefore exactly a weighted harmonic coincidence count**, with harmonics weighted by their own prime complexity. This confirms the physical intuition behind Euler's formula but reveals its self-referential structure: recovering Gradus from coincidences requires weights that are themselves Gradus-like.

Unweighted coincidence counting (flat weights) achieves $\rho = 0.791$ across all coprime ratios with $p, q \leq 16$. The weight function $\log n$ achieves $\rho = 0.795$ without any prime structure, establishing a ceiling for physics-based models that do not invoke prime factorisation.

# 3 Galileo's Model and $\max(p, q)$

Galileo (1638) described consonance in terms of pulse coincidences: two strings at ratio $p : q$ produce coinciding pulses every $p$ vibrations of the lower string. The "sweetness" of the interval grows with the frequency of coincidences, i.e. inversely with $p$. The metric $\max(p, q) = p$ is therefore:

1. **Galileo's pulse model** — vibrations between coincidences;
2. **Harmonic series position** — the upper note is partial $p$ of the implied fundamental [4];
3. **Farey index** — the smallest $n$ such that $q/p$ appears in the Farey sequence $F_n$ (Cauchy 1816);
4. **First coincidence harmonic** — the position in the lower note's series where the two notes first meet.

These four descriptions, discovered independently across four centuries, are mathematically identical. $\max(p, q)$ achieves $\rho = 0.989$ against human ratings, marginally exceeding Gradus, but cannot distinguish intervals with the same numerator (e.g. $5/4$ and $5/3$ both have $\max = 5$).

# 4 An Improved Formula

Euler's formula is symmetric: $\Omega^*(p) + \Omega^*(q)$. Human ratings, however, distinguish numerator and denominator — the bass note and the upper note play different perceptual roles. We find empirically that the asymmetric formula:

$$\boxed{f(p/q) = p + \Omega^*(q)}$$

achieves $\rho = 0.989$, matching $\max(p, q)$ and exceeding Gradus. The formula differs from Gradus only in replacing $\Omega^*(p)$ with $p$ for the numerator. Optimising prime weights freely with six parameters (separate weights for primes 2, 3, 5 in numerator and denominator) achieves perfect rank correlation $\rho = 1.000$, confirming that the structure required to perfectly predict the human rankings is present in an asymmetric prime-weighted formula.

The formula $f(p/q) = p + \Omega^*(q)$ decomposes as the sum of two quantities:

| Component | Formula | Meaning |
|---|---|---|
| Upper note reach | $p$ | Partial number of upper note in implied harmonic series |
| Bass context depth | $\Omega^*(q)$ | Prime complexity of lower note's position in series |

# 5 A Two-Stage Perceptual Model

The formula motivates a two-stage model of how the auditory system evaluates an interval:

**Stage 1 — Bass establishes harmonic context (cost $\Omega^*(q)$).** The lower note is the $q$-th partial of an implied fundamental $F$. The brain extrapolates downward to $F$ by "dividing out" the prime factors of $q$. Each factor of 2 (an octave) costs 1 unit; each factor of 3 costs 2; each factor of 5 costs 4. This is exactly $\Omega^*(q)$. Intervals whose bass note is a power of 2 (e.g. the octave, $q = 1$; the major second bass, $q = 8 = 2^3$) establish the fundamental cheaply via octave equivalence. Intervals with an odd prime in $q$ (the fourth, $q = 3$; the minor third bass, $q = 5$) require the brain to extrapolate through a genuinely new prime, incurring higher cost.

**Stage 2 — Upper note reaches into the series (cost $p$).** Once the fundamental is established, the upper note must be recognised as its $p$-th partial. Higher partials are quieter (amplitude $\propto 1/p$ in typical harmonic sounds), more narrowly tuned, more sensitive to mistuning, and have lower pitch salience [6]. The cost of confirming the upper note as a member of the harmonic series is therefore $p$ itself.

**Total perceptual effort:** $f(p/q) = p + \Omega^*(q)$.

This account connects:

- **Galileo** (1638): pulse coincidences as physical mechanism;
- **Rameau** (1722): the *corps sonore* (harmonic series) as the source of musical meaning;
- **Euler** (1739): prime arithmetic as the language of harmonic complexity;
- **Terhardt** (1979): virtual pitch and harmonic template matching as auditory mechanisms;
- **Huron** (2001): statistical learning of harmonic patterns as the basis of consonance judgments.

## 6 The Remaining Failure

No simple formula distinguishes the major third ($5/4$, human rank 5) from the major sixth ($5/3$, human rank 7). Both have $p = 5$ and $\Omega^*(q) = 2$ (since $\Omega^*(4) = \Omega^*(2^2) = 2$ and $\Omega^*(3) = 2$). The distinction requires knowing that $q = 4 = 2^2$ is simpler than $q = 3$ under octave equivalence, yet the prime-weighting function cannot see this.

Lahdelma & Eerola [8] decompose consonance ratings into sensory and familiarity components; on *sensory consonance* alone, $5/4$ and $5/3$ are nearly indistinguishable across listeners, suggesting that the tie in $f(p/q)$ may accurately reflect the purely arithmetic component of the percept. The gap observed in Krumhansl's data is plausibly attributable to learned exposure — major thirds occur far more frequently than major sixths in tonal music — consistent with the finding of McDermott et al. [9] that consonance preferences in a culture with minimal Western music exposure are substantially reduced. Resolving the tie within an arithmetic formula would therefore require modelling cultural familiarity, which is outside the scope of the present approach.

## 7 Summary

| Formula | $\rho$ vs. human | Notes |
|---|---|---|
| Euler's Gradus: $1 + \Omega^*(p) + \Omega^*(q)$ | 0.979 | Symmetric; two ties unresolved |
| Galileo / $\max(p, q)$ | 0.989 | Simplest; same tie |
| $f(p/q) = p + \Omega^*(q)$ | **0.989** | Asymmetric; two-stage interpretation |
| Optimised asymmetric weights | 1.000 | Six parameters; no closed form |
| Roughness (Plomp–Levelt) | 0.709–0.863 | Register-dependent; fails on tritone* |

* Helmholtz [3] introduced the beating-based dissonance curve that Plomp & Levelt [5] later parametrised from listener data; the two models produce equivalent rankings for our purposes.

Table 1 shows all three formulae alongside the human consonance rank for the 13 standard just intervals of Western music, ordered by human rating. Ties in $f(p/q)$ and in $\max(p, q)$ are indicated by matching values; the sole unresolved tie (major third / major sixth, both $f = 7$) is discussed in Section 6.

Table 1: Dissonance values for the 13 standard just intervals, ordered by human consonance rank [7] (rank 1 = most consonant).

| Interval | Ratio | G$(p/q)$ | $f(p/q)$ | $\max(p, q)$ | Human rank |
|---|---|---|---|---|---|
| Unison | $1/1$ | 1 | 1 | 1 | 1 |
| Octave | $2/1$ | 2 | 2 | 2 | 2 |
| Fifth | $3/2$ | 4 | 4 | 3 | 3 |
| Fourth | $4/3$ | 5 | 6 | 4 | 4 |
| Major third | $5/4$ | 7 | 7 | 5 | 5 |
| Minor third | $6/5$ | 8 | 10 | 6 | 6 |
| Major sixth | $5/3$ | 7 | 7 | 5 | 7 |
| Minor sixth | $8/5$ | 8 | 12 | 8 | 8 |
| Major second | $9/8$ | 8 | 12 | 9 | 9 |
| Minor seventh | $9/5$ | 9 | 13 | 9 | 10 |
| Major seventh | $15/8$ | 10 | 18 | 15 | 11 |
| Minor second | $16/15$ | 11 | 22 | 16 | 12 |
| Tritone | $45/32$ | 14 | 50 | 45 | 13 |

The formula $f(p/q) = p + \Omega^*(q)$ is marginally simpler than Gradus, slightly more accurate, and directly interpretable as a two-stage process of harmonic extrapolation. It suggests that Euler's prime-weighting is correct for the *denominator* but that the *numerator* is better described by the raw partial number — consistent with the auditory system's known sensitivity to partial height rather than prime structure in the upper voice.

## References


[1] L. Euler, *Tentamen novae theoriae musicae*, St. Petersburg, 1739.

[2] G. Galilei, *Discorsi e Dimostrazioni Matematiche*, Leiden, 1638.

[3] H. von Helmholtz, *On the Sensations of Tone as a Physiological Basis for the Theory of Music*, Vieweg, Braunschweig, 1863. (English translation by A. J. Ellis, Longmans, Green & Co., London, 1875.)

[4] J.-P. Rameau, *Traité de l'Harmonie*, Paris, 1722.

[5] R. Plomp and W. J. M. Levelt, Tonal consonance and critical bandwidth, *Journal of the Acoustical Society of America* **38** (1965), 548–560.

[6] E. Terhardt, Calculating virtual pitch, *Hearing Research* **1** (1979), 155–182.

[7] C. L. Krumhansl, *Cognitive Foundations of Musical Pitch*, Oxford University Press, 1990.

[8] I. Lahdelma and T. Eerola, Single chords convey distinct emotional qualities to both naïve and experienced listeners, *Psychology of Music* **44** (2016), 37–54.

[9] J. H. McDermott, A. F. Schultz, E. A. Undurraga and R. A. Godoy, Indifference to dissonance in native Amazonian, *Nature* **535** (2016), 547–550.

[10] D. Huron, Tone and voice: a derivation of the rules of voice-leading from perceptual principles, *Music Perception* **19** (2001), 1–64.

[11] D. L. Bowling, D. Purves and K. Z. Gill, Vocal similarity predicts the relative attraction of musical chords, *Proceedings of the National Academy of Sciences* **115** (2018), 216–221.